Proxy–True Decoupling under Speed-Reward Distortion in CartPole PPO, with a Reproducible Chained-Task ETL Audit

Anonymous submission

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Abstract

We measure whether a speed-shaped proxy objective decouples from the true objective it is supposed to approximate. On CartPole we run a preregistered grid of PPO trainings in which the per-step reward is speed$^{\alpha}$ for $\alpha \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, five seeds per level, and we score every policy against a ground-truth objective that penalises both falling and speeding. The true score peaks at $\alpha = 0.5$ (444.02, 95% CI $[431.89, 456.15]$) and then collapses: $\alpha = 1.0$ reaches -2.22 (CI $[-4.52, 0.08]$) while the proxy score at that same level is the largest observed over the whole grid (1.38), so the proxy ranking and the true ranking are inverted exactly where the proxy is maximised. Under the preregistered decision rule this is recorded as "decoupling confirmed", with the third condition's literal clause annotated as not satisfied (its intent is met). We additionally audit a chained-task measurement pipeline of 90 tasks and 614 steps: the judge is replay-verified (mismatch = 0 over 614 steps). A first batch (one local model, task rules hidden) lands at floor (5/614 steps); we trace the floor to task design rather than model capability, make the rules public while the truth chain stays validator-only, and rerun: semantic correctness rises off the floor and the error decomposition gains a support surface (cumulative errors dominate; an error-then-recover gradient across families is compatible with re-anchoring from the public prompt, reported as exploratory). Both the null and the design-failure diagnosis are reported as measurements.

1 Introduction

Two failure modes recur when an agent is trained or evaluated against something other than the quantity we actually care about. The first is *proxy failure*: the optimisation target is a proxy for the true goal, and maximising the proxy can move the true goal in the wrong direction [4, 3]. The second is *measurement failure*: the procedure that scores an agent is itself buggy, so what we report is not what the agent did [16, 14]. This draft addresses both failure modes separately and deliberately does not address the third (unfalsifiable claims), which is out of scope for the present measurements [11].

Our contribution is a measurement discipline more than a method. For proxy failure we fix the proxy family before touching the trainer, freeze the decision rule, and report the full six-level curve rather than the single best level. For measurement failure we build a judge whose verdicts can be replayed exactly, and we report the floor effect instead of hiding it behind an aggregate.

What we do not claim. Nothing in this draft is a capability claim. The chained-task experiment shows that the measurement pipeline is sound and that the evaluated model performs at floor; it does not show that any architecture is better or worse at chained reasoning.

Table 1: True score with 95% CI, proxy score, and detachment ratio across the α grid. Six levels, five seeds each. Values are copied verbatim from `c2/results/table1_true_score.csv`.

α	true score (mean)	95% CI	proxy score	detach
0.5	444.0216	$[431.894, 456.149]$	0.6429	0.0
1.0	-2.2205	$[-4.519, 0.081]$	1.3762	1.005
1.5	39.4061	$[31.392, 46.777]$	1.0667	0.9113
2.0	59.4365	$[47.910, 71.084]$	0.8549	0.8661
2.5	21.3227	$[16.043, 30.021]$	1.0641	0.952
3.0	11.341	$[10.459, 12.472]$	0.8108	0.9745

2 Proxy–true decoupling under speed-reward distortion

2.1 Design

The true objective is a linear combination of survival and speed regulation:

$$\text{true} = \text{steps_alive} - \lambda \cdot \max(0,\ |\dot{x}| - v_{\text{limit}}), \tag{1}$$

with $\lambda = 1.0$ and $v_{\text{limit}} = 0.376$ frozen before any run (`prereg/preregistration_c2.md: L50`). The proxy objective trained against is the per-step reward speed$^{\alpha}$, with speed $= |\dot{x}|$, over the grid $\alpha \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, five seeds per level (42–46), 100,000 timesteps, deterministic evaluation over 50 episodes (`c2/results/train_config.json: L2--L19`).

One deviation from the preregistration is declared rather than hidden: the agent is trained on the per-step dense reward speed$^{\alpha}$, while the reported proxy score is mean(speed)$^{\alpha}$. The two are not identical, and the declaration is landed at `train_config.json: L25`. We report both the true score and the proxy score at every level, and the preregistration requires `steps_alive` and `mean_speed` to be reported alongside them.

2.2 Preregistered decision rule

The detachment ratio is

$$\text{detach} = 1 - \frac{\text{true}}{\text{true}_{\max}}, \tag{2}$$

and the three-way rule is: *decoupling confirmed* if the true score drops by at least 15% after the peak *and* the 95% CI of the drop excludes zero; *partially* for a 5–15% drop; *refuted* if the true score is monotonically non-decreasing (`prereg/preregistration_c2.md: L63--L68`). The power analysis caps the test-relevant SD at 44.6 (`L81--L86`); the post-run paired differences are 446.2 ± 15.1 and 432.7 ± 16.0 (`L88--L98`).

2.3 Results

Measurements only:

- The true score peaks at $\alpha = 0.5$ (444.02) and falls to -2.22 at $\alpha = 1.0$, then to 11.34–59.44 for $\alpha \geq 1.5$; the post-peak drop spans 86.6% to 100.5%.
- The proxy score reaches its grid-wide maximum at $\alpha = 1.0$ (1.38), which is exactly the level where the true score collapses to -2.22 (CI upper bound 0.081). At that level the proxy ranking and the true ranking are inverted.

Table 2: Batch 1 (rules hidden): 90 tasks / 614 steps, `llama3.1:8b`. `step_acc` is semantic correctness against the hidden schedule; `parse_ok` is JSON well-formedness; `first_err_median` is the median index of the first erroneous step.

family	step_acc	parse_ok	first_err_median
chain-sort	0.0	0.0	0.0
ret-path	0.0	0.0	0.0
stateful-calc	0.0235	0.0235	0.0

- Under the preregistered rule this batch is recorded as "decoupling confirmed". The third condition is not satisfied literally, only in intent; the adjudication is landed at `docs/HANDOVER.md: §13.11` (the three-condition check and the final report wording are both in that subsection).

2.4 Limitations

Five seeds, one environment (CartPole), one reward family (speed$^{\alpha}$). The measured seed SD of 15–16 is below the preregistered 44.6 cap, so power is approximately 1.0 for *this* criterion only; the claim does not extend to other criteria. The α^* that maximises the true score is left-censored at the grid's left edge (every seed peaks at $\alpha = 0.5$), so its variance is degenerate and no positive statement about α^* stability is made.

3 A replay-verifiable chained-task audit

3.1 Task families and judge

Three families of 30 tasks each (90 total): `stateful-calc` (stateful computation chains), `chain-sort` (chained sorting), and `ret-path` (backtracking path search). The judge is a one-way validator: the ground-truth schedule is read only by the validator and is never visible to the model (`src/validators.py: Validator.check_step` scores each step against `truth_chain[k]`). A fail-closed gate (`src/leak_check.py`) probes each family and fails the whole run if any probe answers with the original ground truth more than a calibrated noise band allows (v3.1: a leak signal must dominate the executor's own guess-and-miss noise, both in absolute count and relative to honest execution). Afterwards `src/audit_replay.py` replays every step and requires mismatch $= 0$.

Two batches are reported. Batch 1 evaluates `llama3.1:8b` at temperature 0 with task rules hidden from the prompt. Batch 2 evaluates `qwen2.5:3b` at temperature 0 after the per-step rules (`ops/sort_rule/ptr/values`) are included in the public prompt while `truth_chain` remains validator-only; the schedule file hash is unchanged across batches, so the judge scores the same ground truths. Choosing local models at temperature 0 is a reproducibility choice, not a claim that the configuration is best or strongest.

3.2 Results: a floor, its diagnosis, and the fix

- The judge pipeline is reproducible in both batches: replaying all 614 steps gives mismatch $= 0$ each time (exit status 0).

Table 3: Batch 2 (rules public): 90 tasks / 614 steps, `qwen2.5:3b`. Same judge, same schedules (file hash unchanged).

family	step_acc	parse_ok	first_err_median
chain-sort	0.1168	0.1168	0.0
ret-path	0.1912	0.1912	0.0
stateful-calc	0.1174	0.1174	1.0

- Batch 1 is at floor: semantic correctness 0/197, 0/204, and 5/213 for the three families (5/614 = 0.8% overall); no task is fully correct (0/90).
- The batch-1 floor is a design property, not a capability measurement: the public fields of each family are insufficient to derive the ground truth (the integer operators, the sort rule, and the pointer/value mapping were never in the prompt), so the tasks are unsolvable by construction. The gate passing at floor is a statement about the probe set, not about the model.
- Batch 2 (rules public, truth chain still validator-only) lifts the floor: 0.1168/0.1912/0.1174 per family, and `first_err_median` moves to 1.0 for `stateful-calc` (some tasks now survive step zero).

3.3 Error decomposition gains a support surface after the fix

[exploratory] The single-vs-cumulative error decomposition is the most original part of the analysis and is *not* preregistered, so it is marked exploratory at the section head. Batch 1 had a floor effect (`first_err_median` 0.0 everywhere): the model erred at step zero, so there were no samples of an agent continuing on a corrupted state, and the decomposition had no support surface (`docs/HANDOVER.md: §13.13`). Batch 2 removes the floor, and the decomposition becomes measurable (`qc/q2_q4_analysis/results.json`): cumulative errors dominate in all three families (154/188 = 82%, 133/174 = 76%, 114/165 = 69% for stateful-calc, chain-sort, ret-path respectively). Error-then-recover transitions occur in 46/614 steps, and the recovery rate orders across families as ret-path 27/165 = 16.4%, chain-sort 15/174 = 8.6%, stateful-calc 4/188 = 2.1%. This ordering is compatible with a re-anchoring account: after the rules became public, a model that loses the state can recompute it from the prompt, and the ease of recomputation ranks the families the same way (pointer re-walk > replay-rotate-and-swap > re-apply every operator). We report this as compatibility evidence only: one weak model, no controlled manipulation, no causal claim.

3.4 Limitations

Judge coverage is limited to the three implemented families. Each batch is one model at one temperature with a single run; all batch-2 numbers, including the decomposition and the recovery gradient, are point estimates and carry the exploratory mark. The short-chain-to-long-chain extrapolation question remains unanalysable even in batch 2: per-k sample sizes are 4–30, neither `chain-sort` nor `ret-path` shows a reliable per-k trend ($R^2 \approx 0$), and the `stateful-calc` leave-out test is degenerate (predicting zero against zero). Larger samples or repeated seeds are required before that question can be answered.

4 Penalty visibility is a null result, and null is a data point

A two-arm experiment (`blind` vs `salient`) × five seeds × 100,k tests whether making the speed penalty visible in the observation changes the outcome (`q5/results/train_config_q5.json`). Both arms land at true ≈ 499.99 with `steps_alive` = 500.0 and mean_speed ≈ 0.107 (`q5/results/summary_q5.csv`). The preregistered read-out records the null and attributes it to option (a)—the penalty has no action surface in this setting—rather than to a behavioural effect (`preregistration_q5.md: L48--L54, L67--L76`). We do not extrapolate to a general claim that penalty visibility is unimportant; the preregistration forbids that reading.

5 Discussion

The proxy curve in Section 2 shows the shape the problem usually has in practice: the proxy is not merely uninformative, it is *confidently* informative at the wrong level, and its most confident level is the one where the true objective is worst. The chained-task audit in Section 3 shows the difference between a measurement that can be replayed and a measurement that can only be asserted: the same pipeline that reproduces perfectly also reveals that the model has no partial-credit behaviour to decompose.

What is next. One item only, per the reporting discipline: the chained-task pipeline needs a rerun with a model that is not at floor, since the two analyses it was built for (error recovery trigger conditions, and short-to-long chain extrapolation) are blocked by the floor rather than by the design.

Disclosure. No capability comparison is made anywhere in this draft. All agent-side numbers describe one local model at one setting. The proxy-side numbers describe one environment and one reward family.

Open items before any submission

- Citations are verified for title/year/venue/DOI but not yet for the `zotero_key` column, which is unfilled for all 17 rows (`tools/reconcile_zotero.py` not run).
- `goodhart1975problems` is cited as the 1984 Macmillan reprint chapter (`pp. 91--121`); the original 1975 conference paper's pagination was not located, and the popular short form "When a measure becomes a target…" is Strathern 1997, not Goodhart.
- The project's own "three cracks" taxonomy has no external counterpart: the closest work (Manheim & Garrabrant 2018) uses four categories. The three cited laws must not be presented as the source of a three-way split.
- The exact wording of the "decoupling confirmed" adjudication must be copied verbatim from `docs/HANDOVER.md: §13.11`.
- The α^* censoring disclosure and the power statement must survive into the camera-ready.

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